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DIETARY INTERVENTION IN OLDER ADULTS WITH EARLY-STAGE ALZHEIMER DEMENTIA: EARLY LESSONS LEARNED

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Abstract: In older adults, an adequate diet depends on their ability to procure and prepare food and eat independently or the availability of dietary assistance when needed. Inadequate food intake or increased nutritional requirements lead to poor nutritional status, which is considered a key determinant of morbidity, increased risk of infection, and mortality in elderly individuals. Weight loss among seniors also heralds increased morbidity and mortality. Dietary behaviour disorders affecting food consumption, nutrition status and maintenance of body weight are common in older adults, and have a substantial impact on nutritional status and quality of life among older adults with Alzheimer Dementia (AD). The Nutrition Intervention Study (NIS) is ongoing. It employs a quasi-experimental pre-post intervention design in physically-well, community-dwelling early stage AD patients aged 70y or older. To date, 34 intervention group patients and 25 control group participants have been recruited with their primary caregivers (CG) from 6 hospital-based memory and geriatric clinics in Montreal. The NIS uses clinical dietetics principles to develop and offer tailored dietary strategies to patients and their CG. This paper reports on the application of dietary intervention strategies in two intervention group participants; one was deemed successful while the other was considered unsuccessful. The report documents challenges encountered in assessing and counselling this clientele, and seeks to explain the outcome of intervention in these patients.

Key words: Elderly adults, Alzheimer's dementia, nutrition, diet, diet counselling.

Introduction

Food habits and nutrient intakes evolve with aging (1). In older adults, an adequate diet depends on their ability to procure and prepare food and eat independently or the availability of dietary assistance when needed (2). Inadequate food intake and/or increased nutritional requirements lead to poor nutritional status, which is considered a key determinant of morbidity, increased risk of infection, and mortality in elderly individuals (3). In a recent review, Gillette- Guyonnet et al. (1) noted that food consumption is frequently decreased as seniors with Alzheimer Disease (AD) become less able to carry out activities of daily living, and this is a particularly critical issue for those living alone.

Weight loss among seniors also heralds increased morbidity and mortality (4). Unintentional weight loss is a common predictor of mortality in this sector of the population (5,6) and is one of the main hallmarks of AD, even in its early stages (7, 8) . While longitudinal studies suggest that dementia is often preceded by weight loss (9), it is also typically observed that AD patients are at continued risk of losing weight and a worsening in their nutritional status, which progresses with the evolution of the disease (10-13). Weight loss and poor nutrition may also contribute to the progression and increased severity of cognitive impairment, decreased autonomy, institutionalisation, and mortality (7).

Nutritional risk may occur in the presence of acute diseases, weight loss, chronic states which compromise independence (mobility problems, depression, cognitive disorders), medications with an effect on appetite, and major lifestage changes such as retirement from the workforce and other transitions in personal and social circumstances (1, 14).

Dietary behaviour disorders affecting food consumption, nutrition status and maintenance of body weight are common in older adults and are compounded by diminished functional abilities and insufficient personal, social or material resources (15). Cognitive decline and dementia also affect eating behaviour, with substantial consequences for nutritional status and quality of life (16-18), particularly as the disease progresses (19). While AD is notable for its heterogeneity of symptom presentation, certain attributes are fairly constant (20, 21): memory loss, which occurs early in the disease, affects the individual's ability to purchase and prepare foods, and affects all food-related behaviour. Many with AD experience temporal confusion, forgetting whether they have eaten or which meal they should eat. Their attention span is often limited and they are easily distracted (22). Even in early stages of AD, apraxia can affect the patient's ability to prepare food or eat with regular utensils. As the disease progresses, sensory and perceptive loss (agnosia) may affect vision and smell which can hamper recognition of food items (23), and cause loss of appetite (24). Recent work conducted among community dwelling seniors in the early stages of AD compared to cognitively-normal, age-matched controls has found that even mild-stage dementia may be associated with increased dietary dependency (25, 26). Other functional eating problems common to older adults (e.g. oral health issues, dysphagia) may further compound those specifically related to AD, even in

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early stages. Such observations were recently substantiated in interviews with caregivers (CG) of family members with dementia (27). Since dietary problems in this clientele rarely occur in isolation, clinicians and CG must deal with a range of eating behaviour problems (27, 28).

However, dietary habits among the elderly are not adequately characterised (1, 29), and even less research has been carried out on dietary behaviours among community-dwelling older adults with dementia (27).

Dietary interventions for seniors with AD

The goal of dietary intervention for individuals with AD is to reduce morbidity, improve quality of life, and keep them in the community as long as possible (29, 30). Strategies for maintenance of adequate nutrition and a stable body weight must consider the patient's cognitive and physical state. Family members of AD patients and health care professionals are called upon to help with activities of daily living and offer emotional support (31), which may be critical to the patient's continued desire and willingness to eat. For example, it has been shown that food intakes are better among AD patients who have a good relationship with their care provider (32) thus contributing to improved weight stability among patients and reinforcing the importance of carers as resource persons when planning nutritional interventions for these patients (16).

In an observational study tracking the natural progression of undernutrition in a cohort of physically-well memory clinic patients in early stages of AD, our group found that suboptimal total food consumption in these patients appeared to occur early in the disease, and once in place, remained a characteristic feature (25). This research provided convincing evidence of the need for nutrition surveillance in early-stage AD patients, as they had lower body weight, energy and protein intakes, and were at greater nutritional risk than cognitivelyhealthy controls. Our ongoing work, the Nutrition Intervention Study (NIS) has built on findings on the evolution of nutrition status in this population to develop and offer individualised dietary strategies to a group of community-dwelling AD patients and their CG. The present report thus has three objectives: it describes the application of dietary intervention strategies in two study participants and documents challenges encountered in assessing and counselling this clientele. It also seeks to provide understanding of what works, what does not, and why.

Methods

Population and recruitment

The NIS is ongoing, and employs a quasi-experimental pre-post intervention design (33) in two groups: intervention and control. Using an eligibility checklist, memory or geriatric clinic patients admitted to clinic in the previous year were identified as potential participants by clinic personnel (nurse, coordinator, or clinic physician) in six hospital-based, university-affiliated memory or geriatric clinics in Montreal. For logistical reasons, three clinics were designated as intervention sites and the others were attributed control status. Clinic staff gave a written document prepared by the study team to potential participants and their caregivers. It explained that the study aimed to follow the course of nutritional parameters (principally diet and body weight) in a group of elderly people with memory problems. Clinic personnel informed the NIS coordinator, a research dietitian, of those who agreed to participate after reading the study information, and the coordinator subsequently contacted these individuals to schedule a meeting. Informed consent was obtained at the baseline meeting prior to enrolling subjects into the study. The subject's main caregiver was also recruited to ensure feasibility of data collection, and to examine the contribution of caregiving to the nutritional status of individuals with AD.

Eligible participants were aged 70+ y, had an active caregiver (usually the patient's spouse or adult child), spoke either French or English, were physically well with no substantial weight loss ($\geq$ 4.5 kg in 6 months or $\geq$ 2.2 kg in 1 month) (34) in the year prior to admission to the clinic, were at an early stage of cognitive impairment due to a diagnosis of probable AD according to DSM-IV criteria (21) and the Mini-Mental State Examination, MMSE (35) $\geq$ 22 or stage 3-4 on the Reisberg Global Deterioration Scale, GDS (36), and able to provide informed consent at recruitment. Eligible patients and their caregivers had to be willing to commit for the study period (approximately 6 months). At admission to the Clinic they had undergone comprehensive medical, neurological, and biological evaluations. Exclusion criteria included class III or more severe heart failure, chronic obstructive pulmonary disease requiring home oxygen therapy or oral steroids, cancer treated by radiation therapy, chemotherapy or surgery in the 5 years prior to enrolment (basal skin cell carcinoma admissible), inflammatory digestive disease, or any other chronic illness or condition including visual or auditory problems affecting communication and likely to interfere with diet or participation in the study. The target sample size was 36 intervention patient/caregiver dyads and 36 control dyads, to provide sufficient numbers at 80% power and alpha = .05 to detect a difference of 5% in weight loss between the two groups. As in our previous work in a similar population (37), recruitment has been a major challenge.

Intervention dyads receive the tailored nutritional intervention, while control subjects continue to receive usual care. Both groups are re-assessed after six months. At the end of the intervention period, the programme impact on CG stress will be assessed in the intervention group.

The study was approved by the Institutional Review Board (IRB) at the Institut universitaire de gériatrie de Montréal (Institut) and by the IRB at each participating hospital.

Data collection

The Study Coordinator reviewed patient charts at the Clinic at entry into (T1) and exit from (T2) the study and extracted relevant information (with permission) to provide biological evidence of nutrition status (patient's medical history, cognitive status (MMSE) (35), most recent standard laboratory analyses done systematically at admission to Clinics (albumin, folate, vitamin B12, TSH, CBC, calcium, electrolytes and creatinine). No additional biological samples were obtained in the course of the NIS.

Study interviews for both intervention and control participants took approximately 2 to 3 hours. Instruments administered included a core questionnaire (sociodemographic, general health information, medication use, health perception, and physical activity), functional status (ADL, IADL) (38), the Visual Analogue Scales (VAS) for hunger and appetite from the modified FEED questionnaire (39). Selected anthropometric and physical function measures were obtained using a standardized approach by the trained study coordinator, with participants dressed in street clothes, without shoes. These included weight (beam balance) and height (stadiometer), and grip strength (Martin Dynanometer vigorimeter, Tuttlingen, Germany) (40). Weight history was assessed by an adaptation of the Silhouette method (41); nutrition risk was assessed by the Elderly Nutritional Screening tool (ENS©) (42, 43).

Information on usual diet was obtained by a validated 78-item intervieweradministered FFQ (44) with the caregiver as proxy. Current diet was assessed by two nonconsecutive 24 hour diet recalls or diet records collected by telephone from patients/caregivers in the month following baseline/exit assessment, using the US Department of Agriculture five-step Multiple Pass Method (45, 46). Detailed instructions and supporting documents were distributed to patients and caregivers for reporting diet. Use of nutritional supplements (protein-energy and/or multivitamin-mineral preparations), holistic or herbal preparations, modified food textures and consistencies, therapeutic diets and general health status is recorded. To collect valid dietary data and other information relevant to dietary behaviour and nutritional status from cognitively compromised patients, caregivers were actively involved, and a flexible approach was used within the constraints of patients and their caregivers, where the patient provided data independently; data were obtained by assisted interview; the caregiver replied for the patient; or when the patient had been ill or was hospitalized, or could no longer come to the memory clinic or research center for interviews, the interviewer conducted a home or hospital visit and observed the patient, food-preparation, and/or eating abilities.

Data collection was carried out by the study coordinator at the memory or geriatric clinic where the patient was recruited, with the assistance of a second research dietitian working from the main study site on the telephone-based collection of food records and diet recalls. Both bilingual (French, English) staff members conducted interviews in the participants' language of choice.

Nutrition intervention

The NIS programme was designed using clinical dietetics principles to help the AD patient eat well, to prevent weight loss or maintain his/her healthy body weight (47). Findings from study questionnaires and dietary and nutrition analyses are integrated into a Nutrition Summary document designed by the research team, drawing on medical and clinical dietetics "SOAP" charting principles. This document is used to develop the dietary intervention strategy and also serves as a tool for composing a formal 'Physician Note' mailed to the patient's physician to keep him/her abreast of our findings. A number of tools and techniques have been used to convey dietary recommendations. These include Canada's Food Guide (CFG) (48), the Tufts Modified Food Pyramid for 70+ Adults (49), recipes and tips for increasing intakes of protein and energy and preferred foods, advice for dealing with eating problems, use of food supplements (e.g. Ensure™, Boost™), and importance of adequate fluid intake and physical activity, among others. Additional resources for nutrition intervention in an elderly clientele were drawn from the Ordre professionnel des diététistes du Québec (50), the Gerontology Network of Dietitians of Canada (DC), and the DC decision support service Practice-based Evidence in Nutrition (PEN) (http://www.dietitiansatwork.com/, Accessed 30-04-2008) as well as nutrition education resources such as those developed by DC in collaboration with Professor H. Keller (http://www.dietitians.ca/seniors Accessed 30-04-2008).

The intervention was explained to the patient/CG dyad over the telephone. It was tailored to the individual and based on three to four key objectives for judicious food selection. Patients were encouraged to consume high-quality foods providing adequate energy, micronutrients, beneficial fatty acids, and dietary fibre, to foster good general health and maximise cognitive health (1). Those with good diets were monitored for signs of change to permit rapid reaction and intervention (16, 50). After the 15- to 30-minute phonecall, participant-specific materials were mailed to the dyad. These included results of nutrition analysis in lay terms, our plasticised "Dietary Advice" document, with magnets for affixing to the participant's refrigerator, and additional printed dietary resource information, specific to the participant's needs. Body weight was self- (or proxy-) monitored over the intervention period (51). To ensure accuracy, home scales were verified and calibrated when the participant/caregiver attended interviews, or portable scales were loaned as needed.

Intervention participants and their CG were telephoned at 3-week intervals over the 6-month intervention period to review the participant's eating pattern and provide appropriate counselling. They were also encouraged to contact us as needed. The CG had access to telephone support of the "Dial-a-Dietitian" type, with a call-back service staffed by the study dietitians. A link was also placed on the research centre's website to facilitate CG access to personnel, and point to additional study materials supporting the dietary intervention for those who prefer this mode of communication. The programme also stressed the importance of communication with physicians, dietitians, and other health care professionals, when needed. In the control settings, participants and their caregivers continued to receive the usual evaluation and care provided to patients attending memory and geriatric clinics.

Data handling

Sociodemographic, anthropometric, functional, biological parameters, and ancillary data were entered into Microsoft Excel© files as data were collected. The 24 hour diet recalls were analysed using CANDAT (© Godin, London On), based on the 2007b CanadianNutrient File (CNF2007b) (52), with CANDAT result files output to Microsoft Excel© and exported to SPSS 15 (SPSS Inc., Chicago, IL) or SAS 9 (SAS Institute Inc., Cary, NC). The FFQ was entered using Microsoft Access© software for customised data entry and analysis, with algorithms developed by our study team to compute energy, nutrient and food group values from the instrument, also based on the CNF2007b incorporated into the FFQ data entry utility (44). Adequacy of participants' nutrient intakes was assessed from nutrient recommendations (DRIs) for people aged 70+ y http://www.iom.edu/?id=21372&redirect=0 (IOM, Accessed 30-04-2008). Body weight change over time was examined as percent initial weight (53). Repeated measurements of body weight (and determination of BMI) provided additional evidence of quantitative adequacy of food intakes. Nutrition risk was considered present if ENS© score was ≥ 3 or if weight loss is ≥ 5% initial body weight in the 6-month period, or both.

Results

Table 1 provides descriptive characteristics of NIS intervention and control participants in the study at the time of writing. The sample is currently composed of 34 intervention dyads (patients and caregivers) and 25 control dyads, with mean age situated between 79 and 80 y (range 69 to 90y). There are proportionally more women in the intervention patient group although the difference is not statistically significant, and the two groups are otherwise similar. Both groups were assessed at moderate nutrition risk. At around 25, average BMI in the study sample is in the "healthy" zone, although it ranged from 16.4 to 37.2. At recruitment, most (76% of intervention patients and 96% of controls) were living at home, alone or with family. The majority (59% of intervention participants and 48% of controls) considered their health to be very good.

The two intervention group cases selected for presentation are presented in Figures 1 (Case 1) and 2 (Case 2). At baseline, both had a low BMI; for Case 1 (Mrs. R.) it was 20.8 and for Case 2 (Mrs. M.) it was 16.5. Both participants reported having a poor appetite, which was corroborated by their respective caregivers.

Table 1
Characteristics of Nutrition Intervention Study (NIS) patient sample recruited to date*, at baseline interview

Attributes	Patient Group Intervention (n=34)	Control (n=25)
Gender (n %)		
Male	11 (23)	14 (56)
Female	23 (67)	11 (44)
	Mean ± SD	
Age (years)	79.2 ± 4.9	80.4 ± 5.1
MMSE[1] (/30)	25.1 ± 1.9	24.5 ± 3.2
ENS©[2] (/13)	3.4 ± 1.7	3.4 ± 1.4
Body weight (kg)	64.0 ± 14.5	66.5 ± 19.6
BMI[3] (kg/m^2)	25.1 ± 4.3	25.4 ± 4.2
Grip strength (KPa)	56.7 ± 17.5	55.3 ± 18.7
	n (%)	
Living situation		
Alone	6 (18)	1
With other(s)	26 (76)	24 (96)
Seniors' residence	--	--
Convent	2 (6)	--
Education		
Primary	2 (6)	5 (20)
Secondary	16 (47)	8 (32)
College	5 (15)	5 (20)
University	11 (32)	7 (28)
Self-assessed health at T1		
Excellent	4 (12)	6 (24)
Very good	20 (59)	12 (48)
Good	8 (23.5)	7 (28)
Fair	2 (6)	--

*April 2008; 1. Mini-Mental State Examination (Folstein et al., 1975); 2. Elderly Nutritional Screening© (Payette et al., 1999); 3. Body Mass Index

Figure 1
Case study 1, Nutrition Intervention Study (NIS)

JR is a 76-year-old woman diagnosed with probable AD in May 2007. She has a history of hypertension, dyslipidemia, coronary artery disease, sleep apnea, and hypothyroidism, and suffered episodes of depression at retirement. At the baseline NIS assessment (early June 2007), it was noted that Mrs. R. was apathetic and anergic; her spouse informed us that she slept during the day. She takes a number of medications to control her health conditions, as well as nutrient supplements commonly-prescribed to elderly adults, such as calcium and vitamin D, and she has vitamin B12 injections monthly. Mrs. R. lives with her husband who is her main caregiver.

Mrs. R. is rather tall, measuring 1.70 m. However, her weight of 59.7 kg at the baseline NIS assessment resulted in a BMI of 20.8, well below the recommended BMI for older adults of 22 to 29. In the 2 years prior to interview Mrs. R. steadily lost weight. For example, her medical chart notes that in February 2005, her weight was 70.5 kg, in November 2006, she weighed 63.4 kg, and in May 2007 when she was referred to us, her weight was 62 kg, indicating a loss of 2.2% of her body weight in 6 months. This had further decreased to 59.7 kg in June 2007, when we first met her.

The patient still enjoyed preparing the grocery list and Mr. R. did the shopping. While the caregiver informed us that his wife never wanted to eat, and she reported having a poor appetite - scoring 3/10 on the modified FEED scale (39) - we learned that she enjoyed eating out and that her daughters took her to restaurants several times each week. The only food that she really liked was chicken, which she said she could eat every day. She also reported liking desserts such as ice cream, pastry, chocolate and cookies. Her dislikes included nuts and peanut butter and several vegetables, such as broccoli, cauliflower, green beans and peppers.

Figure 2
Case study 2, Nutrition Intervention Study (NIS)

AM is an 82-year-old woman diagnosed with probable AD in August 2007. Her medical history revealed problems of chronic constipation, intestinal polyps which were surgically removed in 2003, and vascular cardiomyopathy. She is on Reminyl to control her AD and medication to control her angina. Like many older adults, she takes a standard multivitamin once a day and a calcium-vitamin D supplement 3 times a day. Mrs. M. lives with her sister who is her main caregiver.

Mrs. M.'s height is 1.66 m and she weighed 45.5 kg at the baseline NIS assessment (September 2007). Her BMI of 16.5 was well below the recommended BMI for older adults of 22 to 29. According to her sister, Mrs. M had a stable weight for many years (around 55 kg). Episodes of diarrhea and constipation secondary to intestinal polyps, and surgery to remove the polyps resulted in a weight loss of 9 kg which she never regained. Mrs. M. also mentioned that she had lost more weight since she started taking Reminyl. The patient was sedentary, complained of lack of energy, and reported having no appetite; her score of 1/10 on the modified FEED scale (39) was indicative of early satiety. However, according to her sister she did not skip meals. Her score of 7/13 on the ENS© (43) attested to high nutritional risk.

The patient's sister planned and prepared meals and did the grocery shopping. She was devoted to helping her sister avoid weight loss and improve her dietary status; for example, the caregiver added whey protein powder to her sister's soup and cream to desserts, and gave her Ensure Plus™ (235 ml provides 350 kcal and 13 g protein) every day.

Case 1

Dietary assessment revealed that Mrs. R. (Case 1) had intakes falling well under recommendations for all food groups, which was corroborated by nutrient analyses showing inadequate intakes of energy, protein, most micronutrients, and dietary fibre (Table 2). The patient's nutritional risk score, determined from the ENS© (44) was 5/13, denoting moderate nutritional risk. The nutrition intervention plan developed for Mrs. R. is shown in Figure 3. Briefly, we concentrated on preventing protein-energy malnutrition; we recommended that Mrs. R. have a source of protein at each meal and that she adopt a regular meal schedule. We also suggested that she be weighed weekly to monitor her body weight. The nutritional intervention plan was discussed at length with Mr. R, as involvement of the patient's husband was clearly essential. While he expressed his willingness to do whatever was required to help improve her desire to eat, he initially was quite discouraged.

A simplified version of the recommendations was prepared as the NIS "dietary advice" tool, which was plasticised and fitted with magnets so Mrs. R. could put it on her refrigerator. A list of nutrient-dense frozen foods was sent to Mr. R; each of these items had more than 15g of protein per serving, less than 12g of fat (of which fewer than 3g were saturated or trans fat), and under 650mg of sodium. We also recommended that he try to interest his wife in taking short walks (around 5 to 10 minutes per day) to increase her appetite, although we realised that her apathy could be an obstacle to success. A note detailing our findings was sent to the patient's physician and was filed in her medical chart at the memory clinic. Mrs. R's substantial weight gain and her renewed pleasure in eating (Figure 4 and Table 3), along with her caregiver's obvious satisfaction with his wife's dietary improvement led us to qualify this case as a successful nutritional intervention.

Table 2
Case study 1, Nutrition Intervention Study (NIS): Dietary and nutrient intakes and nutrition risk

Food groups and selected and selected nutrient intakes	Patient's intake versus recommendations[1] Patient	Recommendations
Grain products (servings/day)	2	6-7
Vegetables (servings/day)	2.3	3+
Fruit (servings/day)	1.5	2+
Dairy products (servings/day)	0.3	3+
Meat and alternates (servings/day)	0.7	2+
Energy (kcal)	1200	1600
Protein (g)	29	47
Folic acid (ug)	161	400
Vitamin D (IU)	2.0	20
Iron (mg)	6.1	8
Calcium (mg)	251	1200
Dietary fibre (g)	10.5	21
Nutrition risk (/13)[2]	5	--

1. Canada's Food Guide, women aged 70+ y (48); Dietary Reference Intakes [http://www.iom.edu/?id=21372&redirect=0 (IOM, Accessed 30-04-2008)], women aged 70+y; 2. Elderly Nutritional Screening (ENS©, 43); 0-2=no risk. 3-5=moderate risk; 6+=high risk.

Figure 3
Case study, 1 Nutrition Intervention Study (NIS) - Nutrition intervention plan for Mrs. R

The following suggestions were made to the patient and her caregiver:
- Encourage consumption of preferred foods
- Have a source of protein at each meal
  • Chicken as often as possible
  • Try other meat, such as beefsteak, calves' liver, pork, ham…
  • Eat cheese with her toast at breakfast
- Go out to restaurants or eat with her children as often as possible (Mr. R. informed us that they could afford to do this)
- Eat rice more often (the family adds eggs to rice which augments the protein content)
- Have desserts/snacks as long as they don't replace meals:
  • Continue eating ice cream
  • Try good-quality granola bars (specific types and brand names sent to husband)
  • Try "Ensure-type" puddings again, even if they didn't work in the past
  • Introduce certain preferred foods that Mrs. R. has forgotten, for example, cheese with soda crackers.

Case 2

Figure 2 shows that dietary assessment in Case 2 revealed that Mrs. M's intakes fell below recommendations for certain food groups, which was corroborated by nutrient analyses showing inadequate intakes of a number of micronutrients and dietary fibre (Table 4). However, it was surprising that nutrient analyses revealed energy and protein intakes that were more than adequate.

Figure 4

Case study 1, Nutrition Intervention Study (NIS) – Notes from patient/CG follow-up

1- 2007/08/23
- Wt: 61.4 kg (gained 1.7 kg since last visit) – caregiver (husband) bought a new scale
- Husband noticed increased appetite; intervals between eating have decreased to 5 hours and protein consumption has foods/snacks, eats at regular hours increased.
- Plan: emphasis will be kept on protein, eat out with family as much as possible (patient eats more on these occasions).

2- 2007/10/19
- Wt: 63.6 kg (gained 3.9 kg since last visit)
- Husband noticed small decrease in Mrs. R's appetite recently; however, patient is still following the recommendations and eats protein-rich and eats out with family.
- Plan: send tips on how to increase appetite, provide positive reinforcement for husband and continue emphasis on protein.

3- 2007/11/19
- Wt: 65.0 kg (gained 5.3 kg since last visit)
- According to husband, patient's appetite has returned and she eats protein-rich foods and snacks. Husband is very happy about his wife's improved eating habits.
- Husband says that physician decided to stop calcium supplements – did not specify reason.
- Plan: send short list of calcium-rich foods that the patient likes and continue with the other recommendations.

4- 2008/01/10
- Wt: 65.9 kg (gained 6.2 kg since last visit)
- Husband confirms that the patient is drinking calcium-fortified orange juice on a daily basis, has increased cheese consumption, and eats cereal with milk in the morning.
- Plan: encourage patient and caregiver to continue the great work.

Table 3

Case study 1, Nutrition Intervention Study (NIS) - Body weight and BMI over study period

Parameter	Study period: Pre-intervention	Post-intervention
Body weight (kg)	59.7	65.9
BMI [weight (kg)/height (m2)]	20.8	24.1

Table 4

Case study 2, Nutrition Intervention Study (NIS) - Dietary and nutrient intakes and nutrition risk

Food groups and selected nutrient intakes	Patient's intake versus recommendations[1]: Patient	Recommendations
Grain products (servings/day)	3.5	6-7
Vegetables (servings/day)	5	3+
Fruit (servings/day)	2.5	2+
Dairy products (servings/day)	1.3	3+
Meat and alternates (servings/day)	3.5	2+
Energy (kcal)	1857	1400
Protein (g)	81	46
Folic acid (ug)	318	400
Vitamin E (mg)	1.4	15
Vitamin D (IU)	105	600-800
Calcium (mg)	786	1200
Dietary fibre (g)	15.5	21
Nutrition risk (/13)[2]	7	--

1. Canada's Food Guide, women aged 70+ y (48); Dietary Reference Intakes [http://www.iom.edu/?id=21372&redirect=0 (IOM, Accessed 30-04-2008)], women aged 70+y; 2. Elderly Nutritional Screening (ENS©, 43); 0-2=no risk. 3-5=moderate risk; 6+=high risk.

Our nutritional intervention targeted all food groups (Figure 5). We made specific recommendations to increase Mrs. M's intakes of fibre, folate and vitamin E. Intakes of calcium and vitamin D were met by her dietary supplement. While dietary supplements supplied additional nutrients to satisfy the patient's micronutrient needs, we recommended that efforts be made to help the patient improve her intakes through dietary sources. We suggested that she eat several small, high nutrient density meals per day to avoid early satiety and improve her appetite, and provided tips for increasing vitamin E and high fibre foods. We also suggested that she be weighed frequently to monitor her body weight. The patient's sister was contacted by telephone every 3 to 4 weeks to discuss her sister's progress and to learn whether the intervention objectives were met. The recommendations were simplified as the NIS plasticised "dietary advice" tool, fitted with magnets for the refrigerator, and sent by mail to Mrs. M. and her sister. A note detailing our findings was sent to the patient's physician and was filed in her medical chart. We also recommended an appetite stimulant.

Figure 5

Case study 2, Nutrition Intervention Study (NIS): Nutrition intervention plan for Mrs. M

The following suggestions were made, primarily to the caregiver:
- Mrs. M. should be weighed weekly;
- Mrs. M. should eat several small, nutrient- and energy-dense meals daily to reduce early satiety and increase appetite;
- Add vitamin E rich foods to Mrs. M's diet, such as nuts, avocado and olive oil;
- Eat fibre-rich foods, starting with one slice of whole grain bread each day; the issue of hydration will be discussed at the first follow-up;
- Patient should continue taking vitamin-mineral supplements as well as Ensure™;
- Tips for increasing appetite and energy intake will be sent to caregiver ;
- The importance of physical activity and its influence on appetite will be discussed at next follow-up (one month).

Figure 6 shows that several nutritional intervention strategies were attempted with Mrs. M. and that we consistently involved her sister, the caregiver. Despite the caregiver's best efforts, the patient continued to lose weight. Her food intake did not improve and the caregiver became progressively more discouraged. Although the caregiver was determined to stay by her sister until the last mouthful was eaten, meals took increasingly longer with a simple lunch lasting up to 2 hours. Our role was to continue encouraging the caregiver's efforts, particularly as she told us that they were helpful. While Mrs. M's body weight was reasonably stable (Table 5), she had no desire to eat. Despite the caregiver's persistence, she appeared to be at the end of her resources. For these reasons, we considered Mrs. M's case as an unsuccessful nutritional intervention.

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Figure 6

Case study 2, Nutrition Intervention Study (NIS) – Notes on patient/CG follow-up

1- 2007/11/05
- Wt: 45.0 kg (lost 0.5 kg since last visit) .
- Caregiver says that her sister now eats more fibre (multigrain bread), but she can't eat several meals per day (no appetite); still enriches sister's soup with whey protein powder, puts cream in her cereals, and adds butter to her vegetables. She also tries to encourage her sister to drink a whole can of Ensure Plus™ per day (previously ½ can).
- Caregiver mentions that patient prefers soft foods and would like suggestions for good-quality energy bars and shakes that would help her sister eat high calorie foods.
- New medications.
- Plan: Continue weighing the patient regularly, send suggestions on energy bars and shakes/supplements, provide positive reinforcement.

2- 2007/12/10
- Wt: 44.7 kg (lost 0.8 kg since last visit); varies between 43.6 and 45.6 kg.
- Sister says that the patient has no appetite, eats small meals but never finishes them. She adds butter, cream, whey protein powder wherever she can, leaves high energy snacks on the table all day (chips, chocolate, nuts), adds whipped cream to desserts; she also bought or prepared all the products we suggested.
- Patient still shows preference for soft texture foods and soups.
- Plan: Blenderise meals and serve as soup. After ruling out alcohol-medication interaction, we suggested that the caregiver give her sister a small glass of wine before supper to stimulate her appetite. This positive reinforcement addressed the caregiver's obvious state of being overwhelmed and discouraged about her sister's situation.

3- 2008/01/11
- Wt: 44.5 kg (lost 0.2 kg since last visit).
- Caregiver has followed all recommendations (pureed food, glass of wine) but finds no improvement. Even with family and celebrations during the Christmas holidays the patient did not eat better. Patient will occasionally drink ½ can of Ensure™.
- Plan: 1) continue encouraging caregiver's great efforts; 2) keep physician informed of situation.

4- 2008/02/18
- Wt: 44.8 kg (slight gain of 0.3 kg since last visit).
- Patient is now drinking 1 can of Ensure Plus™ per day (with medication) and caregiver offers "hot chocolate" (Carnation breakfast™ enriched with 15% cream) to the patient at least once a day, with whipped cream on top.
- Caregiver follows the patient's moods (when she feels good, she serves more food, etc.) and offers several snacks per day (cookies with whipped cream, Ensure-type puddings)
- Caregiver mentions that the patient has more appetite when eating pasta.
- Plan: Overall goal: maintain a stable weight, prevent weight loss for now; 1) encourage caregiver's efforts; 2) suggest she serve pasta more often and add cream, cheese, or eggs to the dish; 3) patient should always take medications with high energy beverage (Ensure Plus™, etc.); we will send a sample of a 2.0 kcal/mL supplement (Resource™ 2.0) to drink with medications, providing an extra 470 kcal/day.

Table 5

Case study 2, Nutrition Intervention Study (NIS) - Body weight and BMI over study period

	Study period	
Parameter	**Pre-intervention**	**Post-intervention**
Body weight (kg)	45.5	44.8
BMI [weight (kg)/height (m2)]	16.5	16.1

Discussion

It is generally acknowledged that good nutrition promotes optimal functioning and fosters quality of life by maintaining adequate levels of micronutrients and avoiding malnutrition (54). Further, limiting weight loss has positive consequences since screening for dementia among seniors has shown that weight loss is an independent predictor of mortality (55). Since physicians treating older people are not trained to appraise the quality of their patients' diet (56), deterioration in food consumption and nutritional status is often missed among seniors living in the community (57) despite their increased nutritional risk (58). Because nutrition problems may be reversible (59-61), patients in the early stages of Alzheimer's disease (AD) would benefit from dietary assessment and targeted interventions designed to stabilise or improve their nutrition status. Such action could lessen the nutritional consequences of cognitive decline in those already affected by AD, postpone institutionalisation and have a positive impact on public health (14, 16, 62) and CG burden (63).

The nutrition intervention strategies described in this report were designed to target modifiable dietary problems and help AD patients maintain adequate nutritional status (47). In the present paper we have recounted our experience with two study participants and their respective caregivers. As our involvement with these – and other - patients progressed, it became obvious to us that some seniors with AD will benefit from a targeted nutritional intervention, while success will be elusive among others, and in these patients, even the most appropriate intervention and the most dedicated caregiver cannot stop the slide to weight loss and increasing nutrition risk.

Considerable effort was devoted to providing CG with accessible information sources, tips and tools to support them in caring for their family member with AD. While CG were generally receptive to verbal communication and printed material, we found that they preferred direct contact with the study team and rarely accessed either the telephone support line or the website set up for this purpose. This could be due to an age effect, in that most of the CG enrolled in the study were the elderly spouse of the patient (or in the present illustration of Case 2, a sibling of similar age to the AD patient). It could also result from the fact that they were simply overwhelmed by their care burden, and drawing on technology as an information source did not appeal to them. Since there is little in the literature to support or refute these comments, they may be considered somewhat speculative at this time.

Numerous hypotheses have been proposed to explain weight loss in AD, ranging from decreased food consumption and depression (10), to atrophy of the mesial temporal cortex causing disturbances in appetite-related mechanism (12, 65), to increased energy needs due to patients' tendency to wander (8, 58, 66). Weight loss may also be caused or exacerbated by medications used to treat AD (66-69), but their impact on gastrointestinal function and appetite, and consequently on body weight is variable among patients and does not tell the whole story. In addition, there is no clear evidence linking metabolic disturbances to resting metabolic rate (70) or weight loss among AD patients (71). Whatever the cause, weight loss in AD patients depletes both fat tissue and lean body mass (12) and may be interspersed with periods of weight gain (22).

Poor dietary intakes (especially energy and protein) are common in this patient population (25), and a variety of factors - notably lack of interest in meals or refusal to eat - may contribute to poor intakes and weight loss. On the other hand, we were able to show that diet and body weight can be improved in some AD patients as long as food intakes and physical activity remain adequate (10). Our observations have therefore partially substantiated others' findings that early intervention can lead to improvement in nutrition status (1, 13). However, we must stress that intensive dietary intervention in this clientele may lead to results that vary widely from one patient to another. Since success of dietary intervention among community-dwelling seniors with AD will depend in large part on the pivotal role played by the caregiver in ensuring that dietary advice is applied, constant support must be offered to CG to bolster their efforts (63).

This study is not without limitations. We deliberately selected two contrasting cases to illustrate success or failure of the nutrition intervention. Our selection criteria were drawn from clinical dietetics principles which ascribe success to an intervention that leads to positive results in diet and body weight. This may have conferred bias; for example it is possible that the age difference between our cases (Mrs. M. was 6 years older than Mrs. R) contributed largely to "failure" in Case 2, since her greater age alone may have conferred additional nutritional risk. It should be noted that both Cases' CG were very receptive to the intervention and provided exemplary care, so it is unlikely that we can ascribe success or failure to their efforts.

There may have been anomalies in the dietary data collection, such as our assessment of Mrs. M's energy and protein intakes during a stay at the family's home in the countryside. Indeed, it emerged that these intakes probably resulted from circumstances related to the fact that the two food records were collected while the patient and her sister were staying at their country home, where Mrs. M. was more relaxed and more interested in eating. As can be observed in the follow-up notes (Figure 6), in the weeks following our baseline assessment, the sisters returned to the city and Mrs. M's appetite began to deteriorate, resulting in her slide toward weight loss. It is thus possible that her intakes during this time were not representative of her usual food consumption habits since she steadily lost weight over the follow-up period.

Perhaps the NIS follow-up period could have been longer; however, budget constraints precluded extending our patient follow up. No biological data were collected in the NIS; but access to physiological parameters could have shed some light on the discrepancy between dietary intakes and body weight in Case 2. Finally, full analyses on all NIS participants are expected to provide valuable information on the impact of the intervention and its relationship to maintenance of adequate nutritional status.

Conclusions

A number of nutrition intervention studies in AD patients have provided evidence of increased or maintained body weight (72) and better values in a number of clinical and biochemical parameters. Since prevention of weight loss in AD is central to maintaining quality of life for these patients (29), geriatric assessment must include systematic detection of nutritional problems in addition to monitoring body weight (risk of undernutrition, eating behaviour problems), and application of an appropriate nutrition intervention where warranted (73). However, to our knowledge, seniors with cognitive problems followed at memory or geriatric clinics do not routinely undergo dietary assessment. While patients in these settings are regularly weighed, and staff may detect and chart weight fluctuations, dietary assessment and nutritional evaluation is not part of the systematic patient review process. Furthermore, their caregivers lack the information, resources and support needed to help their family members eat adequately. Without timely and effective intervention, nutrition problems may progress to a point where it becomes difficult to reverse the slide. While the present report provides a glimpse into the impact of dietetic intervention in seniors with early-stage AD and more research is required to determine whether AD progression may be slowed by nutritional intervention (7), it would appear that nutrition professionals are much needed in settings where community-dwelling AD patients are treated. As usual practice in geriatric medicine is based on interdisciplinarity and patient-centred care, clinicians must work together in the best interest of the patient.

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